

Dual-Fitness Optimizer Benchmark for Neural Intraday Trend Classification: Methodology and Results

1 Materials and Methods

1.1 Study Design and Data Partitioning

This report evaluates whether the *selection objective* used by an optimizer changes the out-of-sample behaviour of neural and machine-learning classifiers. The data comprise 15,057 chronologically ordered 5-minute bars from B3 Mini-Index futures, obtained from the BovDB collection [1, 2]. Seven technical indicators were retained by Information Gain screening. All experiments use the same sequential 60/20/20 train/validation/test split, with 9,034 training bars, 3,011 validation bars, and 3,012 locked test bars. No shuffling is performed.

Figure 1 makes the experimental comparison explicit. The data, features, model families, optimizer families, and evaluation budget are shared; only the validation fitness changes. The test set remains unavailable during both optimization procedures.

1.2 Shared Search Controls

Four model backbones (MLP, Random Forest, SVM, and 1D-CNN) are optimized by Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO). Each model-optimizer-protocol combination is given the same cap of $N_{\text{eval}} = 1,500$ candidate evaluations per seed. This equal-budget condition prevents an optimizer from obtaining an apparent advantage merely by evaluating more candidates.

1.3 Two Validation Fitness Functions

Protocol A selects candidate configurations using a compound validation objective designed to emphasize imbalance-robust discrimination:

$$f_{\text{MCC/F1}}(\boldsymbol{\theta}) = 0.60 \text{MCC}_{\text{val}}(\boldsymbol{\theta}) + 0.40 F_{1,\text{val}}(\boldsymbol{\theta}). \quad (1)$$

Protocol B uses the conventional validation-accuracy objective:

$$f_{\text{Acc}}(\boldsymbol{\theta}) = \text{Accuracy}_{\text{val}}(\boldsymbol{\theta}). \quad (2)$$

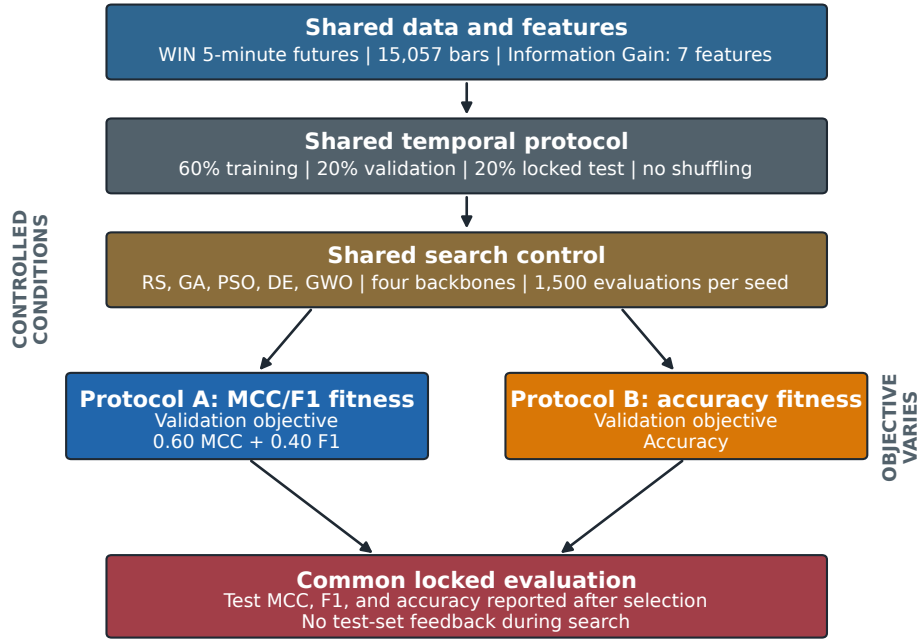


Fig. 1 Controlled comparison of two optimization protocols. Shared conditions are held fixed; the validation objective alone varies. Both protocols are evaluated once on the same locked test block after model selection.

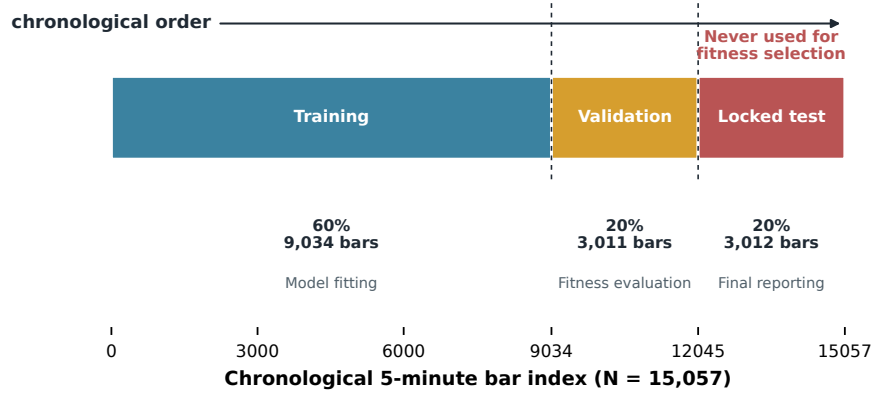


Fig. 2 Chronological train/validation/test partition used by both protocols. The test block is never used for fitness evaluation or hyperparameter selection.

The objective is used solely to rank candidate configurations on the validation block. After selection, both protocols report test accuracy, test MCC, and test F_1 . Consequently, test accuracy under Protocol A is a secondary outcome of MCC/ F_1 -directed selection, whereas test accuracy under Protocol B is the primary outcome of accuracy-directed selection. This distinction is essential: a reported metric and the fitness that generated the selected model are not interchangeable.

1.4 Available Evidence and Reporting Scope

The direct protocol comparison uses the three paired seeds available in both protocols (seeds 1–3) for every one of the 20 model–optimizer cells, yielding 60 seed-level observations per protocol. Figures that contrast the two objectives therefore remain descriptive. A complementary, model-level exploratory analysis treats the five optimizer–seed combinations as 15 matched blocks within each protocol and applies Friedman tests followed by Holm-adjusted paired Wilcoxon contrasts. This analysis compares models *within* a protocol; it does not establish a confirmatory difference between the two fitness functions. Confirmatory inference on the fitness contrast requires a prespecified rerun with at least 20 common seeds per cell.

2 Results

2.1 Objective-Dependent Out-of-Sample Performance

Figure 3 displays the held-out outcomes for each of the 20 model–optimizer cells. Across these cells, the unweighted macro mean test accuracy was 0.623 for MCC/ F_1 fitness and 0.617 for accuracy fitness. The corresponding macro mean test MCC values were 0.247 and 0.235, respectively. These values are descriptive averages of cell-level means, not population estimates.

The paired display is more informative than a single aggregate bar: it shows that the effect of the objective varies across configurations. Accuracy-directed selection does not uniformly increase held-out accuracy in the currently available runs, and its effect on MCC is likewise backbone- and optimizer-dependent. This is precisely the empirical question that the dual-fitness design makes testable once the confirmatory seed budget is complete.

2.2 Where the Objective Changes Performance

Figure 4 resolves the protocol contrast by backbone and optimizer. Each cell reports the signed difference between the accuracy-fitness and MCC/ F_1 -fitness protocols. Positive values favour accuracy fitness and negative values favour MCC/ F_1 fitness. The pattern is heterogeneous rather than universal, which precludes a defensible claim that either objective is globally superior on the present three-seed evidence.

2.3 Convergence Under the Two Objectives

Figure 5 shows search dynamics for the MLP reference backbone. The left panel tracks the compound MCC/ F_1 validation fitness, whereas the right panel tracks validation

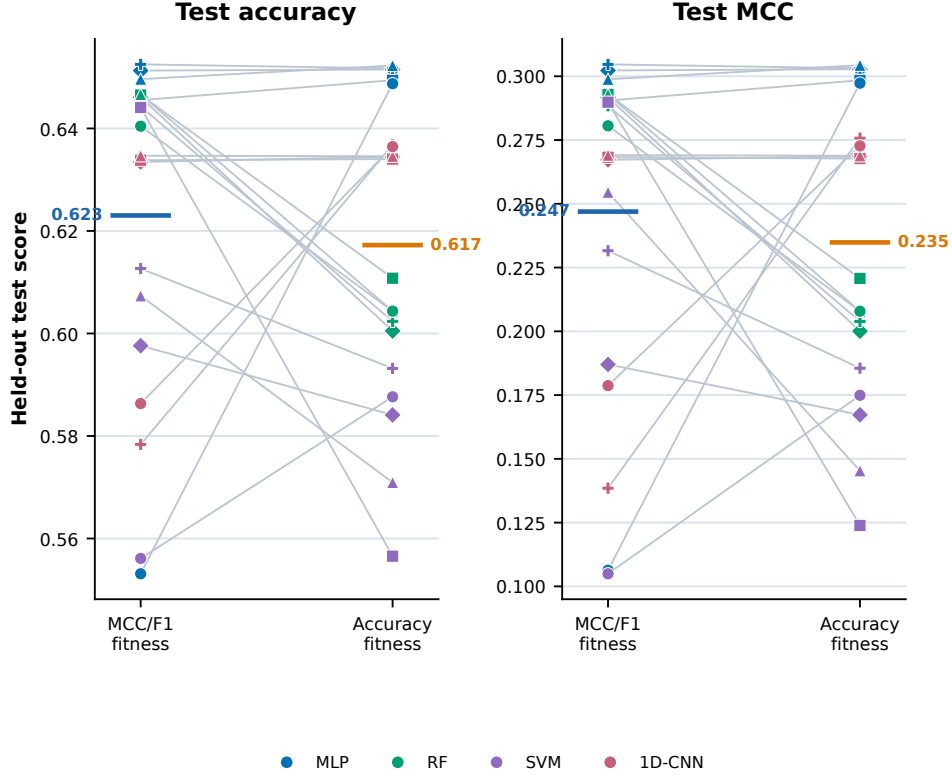


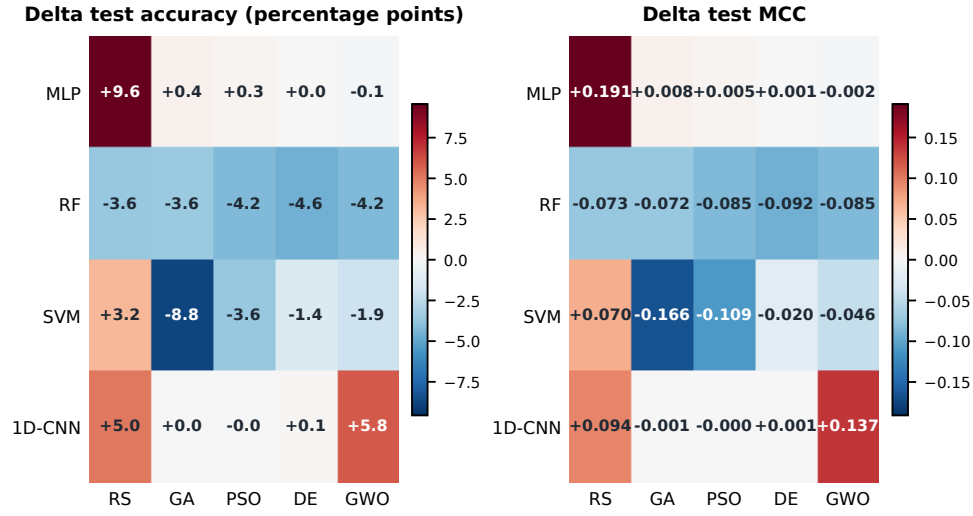
Fig. 3 Held-out performance under the two validation objectives. Each connected pair is the mean of the same three seeds for one model-optimizer cell; colour identifies the backbone and marker shape identifies the optimizer. Short horizontal segments give the macro mean across the 20 cells.

accuracy. These curves must not be interpreted on a common vertical scale: they represent different objectives. Their role is to show how each optimizer progresses under the criterion it was assigned, not to establish cross-protocol statistical superiority.

2.4 Exploratory Statistical Evidence

Figure 6 reports the precomputed model-level tests separately for the two holdout protocols. Each Friedman test compares the four model backbones within 15 matched optimizer-seed blocks. Under MCC/F1 fitness, the test-MCC ranking differs across models ($\chi^2 = 22.84$, $p = 4.36 \times 10^{-5}$). Under accuracy fitness, the test-accuracy ranking also differs across models ($\chi^2 = 42.92$, $p = 2.56 \times 10^{-9}$), with MLP obtaining mean rank 1.00. The Holm-adjusted MLP contrasts in Protocol B are significant against CNN, RF, and SVM ($p_{\text{Holm}} = 0.00194$ for each comparison).

These are useful exploratory checks of model separation under a fixed protocol, but they must not be recast as a direct significance test of MCC/F1 fitness versus accuracy fitness. The three common seeds remain insufficient for that confirmatory claim.



Delta = accuracy-fitness protocol minus MCC/F1-fitness protocol; positive values favour accuracy fitness.

Fig. 4 Configuration-level change in held-out metrics induced by the validation objective. Values are accuracy-fitness minus MCC/F1-fitness means over the paired seeds 1–3. The left panel uses percentage points; the right panel uses raw MCC units.

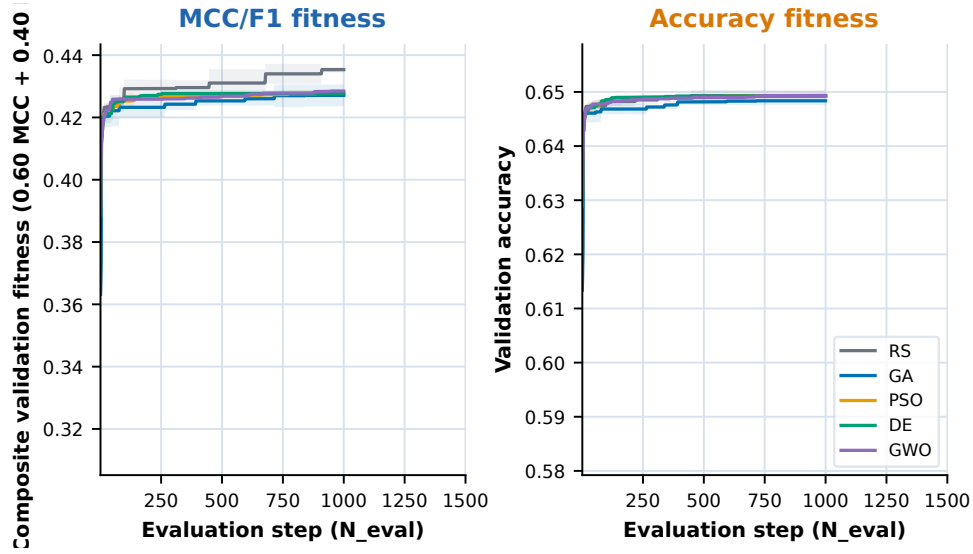


Fig. 5 Convergence trajectories for the MLP reference backbone under the two validation objectives. Solid lines are means and shaded ribbons are ± 1 standard deviation across the three paired seeds. Left: composite MCC/F1 validation fitness. Right: validation accuracy.

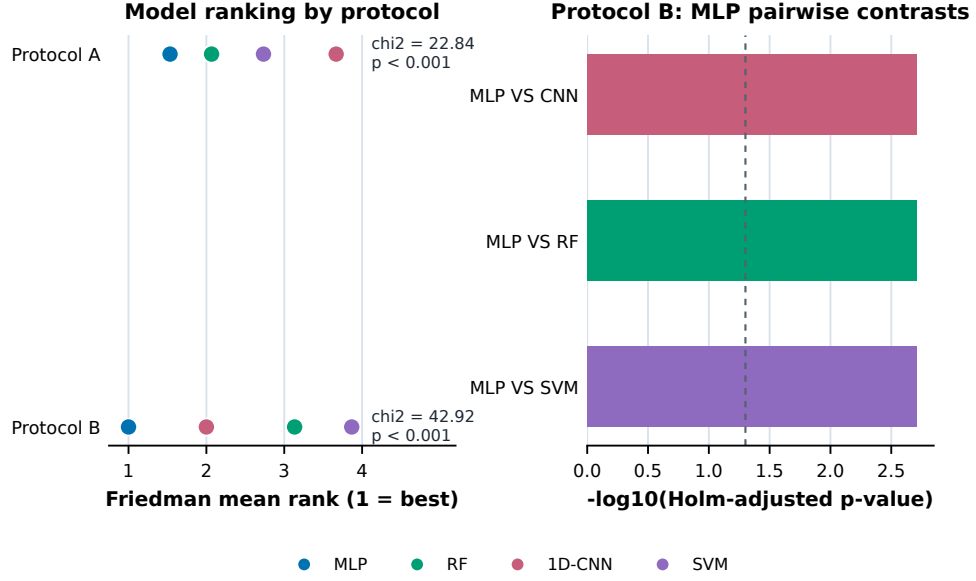


Fig. 6 Exploratory model-level statistical evidence. Left: Friedman mean ranks across 15 matched optimizer-seed blocks per holdout protocol; lower rank is better. Right: Holm-adjusted paired Wilcoxon evidence for MLP against the other models under the accuracy-fitness protocol; the dashed line marks $\alpha = 0.05$.

2.5 Economic Evaluation

The same held-out predictions were assessed by an intraday long/short trading simulation. Figure 7 summarizes total profit and the return-drawdown trade-off over the 15 optimizer-seed blocks for each model and protocol. Under MCC/F1 fitness, RF has the highest mean total profit (26,465 points), but the corresponding Holm-adjusted pairwise profit contrasts are not significant. Under accuracy fitness, MLP has the highest mean total profit (27,737 points); its profit advantage is significant versus RF ($p_{\text{Holm}} = 0.00018$) and SVM ($p_{\text{Holm}} = 0.00305$), but not versus CNN ($p_{\text{Holm}} = 0.169$). RF retains the least negative mean drawdown under accuracy fitness (−1,866 points), illustrating that economic preference depends on the chosen risk criterion rather than on predictive accuracy alone.

2.6 Interpretation and Confirmatory Next Step

The present results establish a controlled, visually auditable comparison between two fitness functions, together with exploratory model-level predictive and economic evidence. The next experimental stage should complete a common 20-seed design for every model-optimizer cell, retain the same temporal split and budget, and apply a prespecified paired analysis separately for test MCC, test accuracy, and economic outcomes. Only then should omnibus and post-hoc tests be used to quantify objective-specific advantages.

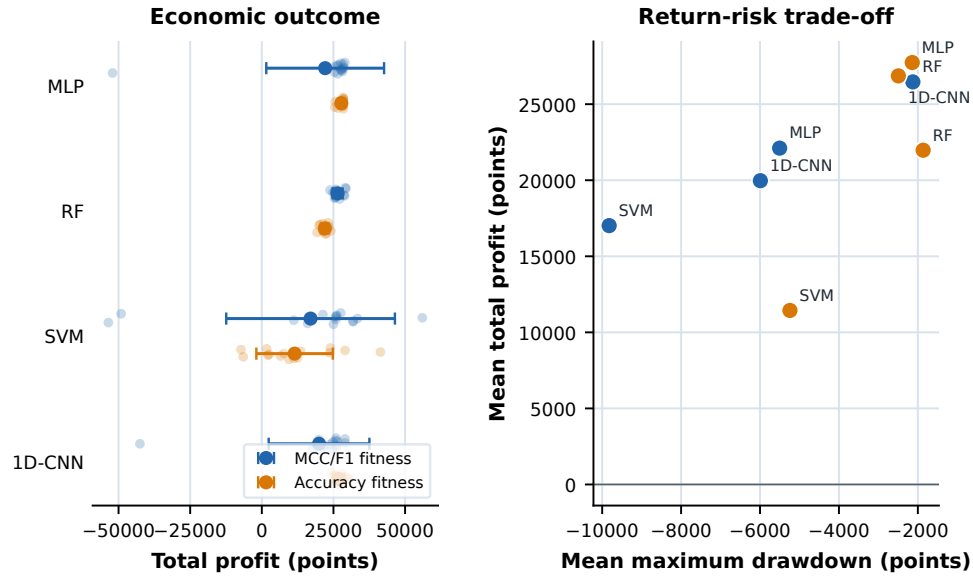


Fig. 7 Exploratory economic evaluation under the two holdout protocols. Left: total-profit observations and mean ± 1 standard deviation across 15 optimizer-seed blocks. Right: mean return versus mean maximum drawdown; values farther right have less severe drawdown.

References

- [1] Cardoso, F.C., Malska, J.A.V., Ramiro, P.J., Lucca, G., Borges, E.N., Mattos, V.L.D., Berri, R.A.: Bovdb: a data set of stock prices of all companies in b3 from 1995 to 2020. *Journal of Information and Data Management* **13**(1) (2022)
- [2] Souza, A.V.S., Lucca, G., Borges, E.N., Cardoso, F.C., Dalmazo, B.L., Berri, R.A.: Dataset for Intraday Analysis of B3 Stock Prices. Harvard Dataverse (2024). <https://doi.org/10.7910/DVN/TMB4IG>